Tunnels and Underground Structures, Zhao, Shirlaw & Krishnan (eds)
© 2000 Taylor & Francis, ISBN 90 5809 171 6

Tunnelling under shop-houses along Serangoon Road, Singapore

R.S.Nair
North East Line Contract 705, Land Transport Authority, Singapore
D.Wen
Land Transport Authority, Singapore

ABSTRACT: This Paper will discuss the construction of the bored tunnels underneath a row of shop-houses on Contract 705 of the North East Line Project (NELP). The operation of Earth Pressure Balanced Machine (EPBMs) during tunnelling will be discussed with detail on the operation parameters of face pressure, grout pressure and grout volume. The ground and building settlement monitoring of the old shop-houses will also be discussed as a means of examining the EPBM's performance.

1 INTRODUCTION

Contract 705 of the NELP includes the construction of two civil defence stations, Boon Keng Station (BNK) and Sennett Station (SNT), cut & cover tunnels and twin bored tunnels. The bored tunnels have an internal diameter of 5.8m and are lined with 250mm thick pre-cast reinforced concrete segmental linings. About 200m of the South Bound (SB) bored tunnel between the two stations has been constructed directly under the old two storey shop-houses along Serangoon Road between Boon Keng Station and Kallang River, see figures 1 and 2. Parallel to the SB tunnel is the North Bound (NB) tunnel with a centre to centre distance of 22m. The distance from the centre of the NB tunnel to the edge of these shop houses ranged from 6 to 13m.

The bored tunnels are typically 21.5m from the ground level to tunnel axis in the Jurong Formation of the Old Alluvium with soft clay above the tunnel crown, Figure 3. Locally the tunnel crown protruded marginally into the soft clay layer. The two storey shop-houses were built during the early 20th century and inspections of these buildings revealed that these were in a poor condition.

Contract 705 was awarded to Kumagai/SembCorp/Mitsui Joint Venture (KSMJV). Two earth pressure balance machines (EPBM) have been used for the drives. The Contract requires the NB EPBM to be kept at 15m ahead of the SB EPBM during construction.

It was agreed that the NB EPBM would commence the tunneling work first along Serangoon Road. Based on the results of the instrumentation and the EPBM operational parameters after the passing of the NB EPBM, KSMJV and NELP would then decide whether it would be safe enough for the SB EPBM to tunnel underneath the shophouses.



Figure 1. Two storey shop-houses above SB bored tunnels of C705.

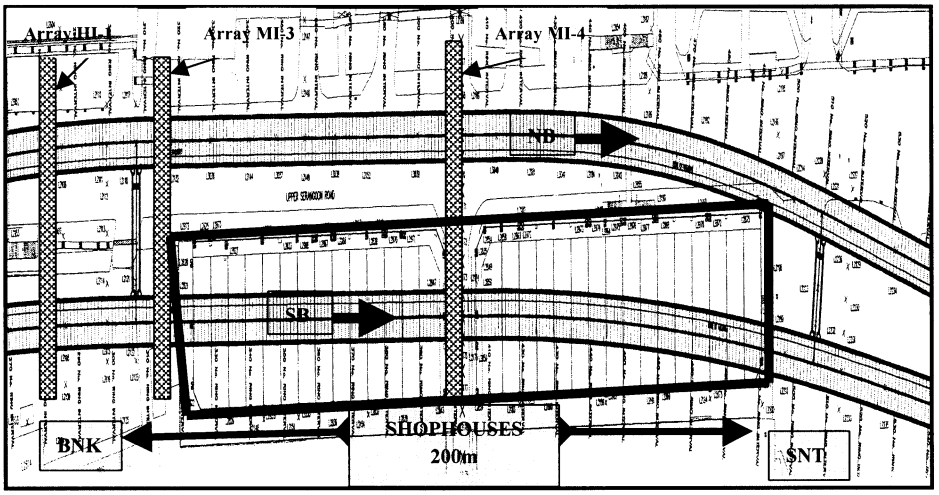
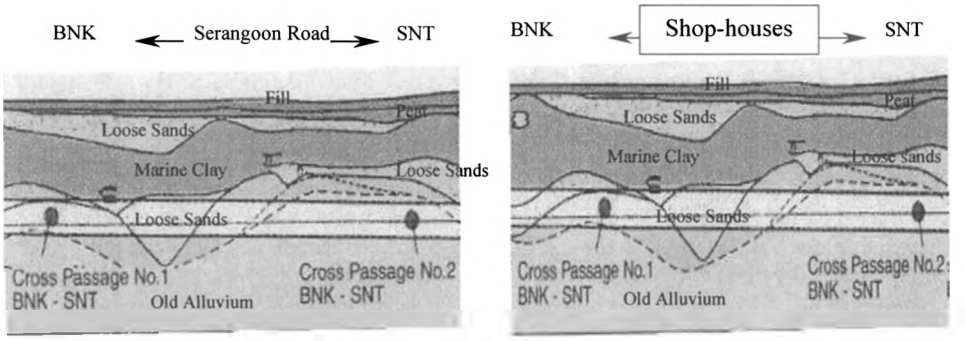


Figure 2. Plan view of tunnel route



NB Longitudinal Soil Profile

SB Longitudinal Soil Profile

Figure 3. Longitudinal soil profiles.

2 INSTRUMENTATION AND MONITORING OF SHOP-HOUSES

2.1 *Proposed instrumentation*

The contract required the contractor to set up an intensive monitoring zone in front of the shop-houses to confirm the performance of the machine before going under the shop-houses. After discussions with NELP, Contractor agreed to install three monitoring arrays for this purpose. These consisted of two medium intensity (MI) and one high intensity (HI) arrays, the details of which are shown in Table 1. Locations of the arrays are shown in Figure 2.

Surface settlement points monitored by precise leveling were installed at spacing not more than 25m along the centre line of each tunnel. Additional surface settlement points are distributed within the predicted 5mm-settlement contour. Building settlements around the shop-houses and at the columns were also installed. Two series of electrical level beams were mounted on the shop-houses, one parallel and one perpendicular to the advance direction of the TBM. Crack-meters were installed at existing cracks in the buildings.

Table 1. Instrument arrays

Array No.	Type of Instruments	Location
High Intensity 1 (HI1)	• Inclinometers • Spider magnet extensometers • Pneumatic piezometers • Settlement points	80m in front of the shop-houses to monitor TBM performances before going underneath the shop-houses
Medium Intensity 3 (MI3)	• Spider magnet extensometers • Pneumatic piezometers • Settlement points	5m in front of the shop-houses to confirm the TBM performances immediately before going underneath the shop-houses
Medium Intensity (MI4)	• Spider magnet extensometers • Pneumatic piezometers • Settlement *points*	Between the two cluster of shop-houses to confirm the TBM performances immediately before going underneath the second cluster of shop-houses

2.2 *Frequency of monitoring*

Intensive 8-hour monitoring array started when the TBMs were 25m away from the array and continued until the EPBM was 10m past the array. These arrays were read at a maximum of 4 times a day. The building settlement and ground settlement points were monitored at every 3 hours when the TBM was within 25m (in front or behind) of each individual shop-house. The crackmeters inside the shop-houses were read twice a day and those outside the shop-houses at every 3 hours.

The shop-house monitoring team was based in the TBM monitoring room. All data was forwarded to the engineer every 3 hours for analysis to ensure that a communication channel was kept open between the monitoring team and the TBM operation team. Data from the monitoring would be analyzed and if necessary the operation of the TBM could be adjusted to minimize the settlement of the ground.

3 CONSTRUCTION

3.1 *Cutterhead Inspections*

Prior to tunnelling underneath the shop-houses, cutterhead inspection for each TBM was carried out to check the wear and tear of the picks. This was done when the machine was 25m away from the edge of the shop-houses. The other parts of the machine were also checked. The cutterhead inspection was necessary so that there would be continuous working during tunnelling operations underneath / adjacent to the shop-houses.

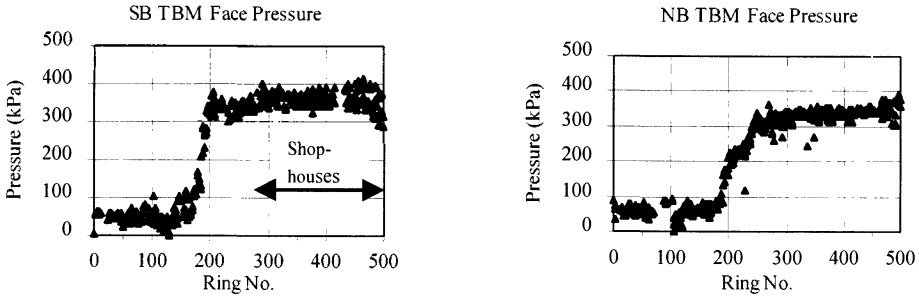


Figure 4. TBM face pressures.

3.2 *TBM face pressures*

The EPBM provides continuous support of the tunnel face using freshly excavated soil, that completely fills the work chamber. The supporting pressure is controlled by adjusting the incoming and out-going soils in the chamber. Excessive pressure on the tunnel excavated face reduces the life span of the TBM's cutter head while insufficient pressure will cause ground surface settlement. In view of the shop-houses directly above the SB tunnel the face pressure was carefully monitored to ensure that the settlement at the ground surface was kept at minimum.

The EPBMs were equipped with three earth pressure sensors. During the cutterhead inspection for both machines at about 25m before the shop-houses, all earth pressure sensors were tested to ensure that they were working properly. The overburden pressure below the shop-houses is in the range of 360 kN/m^2. The contractor proposed to maintain the earth pressures between 320 to 360 kN/m^2 during tunneling directly under the shop-houses, Figure 4. If face pressures dropped during ring erection, sufficient polymer would be injected into the cutterhead to increase the pressure to a minimum of 320 kN/m^2. In normal situations, all the rams would be withdrawn after excavation before commencement of ring erection. However when tunneling underneath the shop-houses the rams were only withdrawn when necessary to erect each segment.

The contractor worked 24 hours, continuously monitoring the face pressures and at the same time checking the settlement at the ground surface until the EPBMs cleared the shophouses. Communication was maintained between the driver in the operating panel of the EPBM who controls the face pressures and the staff who actually monitored the readings at the surface. This was important as it allowed the Contractor to look out for tell-tail signs on the ground surface and the buildings when the EPBM's were beneath the shophouses.

3.3 *Grouting*

Each of the EPBMs had a total of 4 grouting ports, 2 on the top and 2 at the bottom of the face. The design of the machine was to grout the tail void behind the segments automatically during excavation. Unfortunately, after completing about 300m of tunnelling from Boon Keng station, there was constant blockage of the grout ports. In order to prevent unnecessary stoppages of the EPBM underneath the shophouses, it was agreed to allow the Contractor to do manual grouting of the segments through the grout hole provided at the lifting eyes. To maintain continuous grouting throughout the shove, the grouting points on the last ring expelled from the shield were used. This would ensure that the gap around the lining was immediately filled. Trials indicated that equal volumes of grout could be pumped through the grout holes to fill the annulus behind the rings. When tunnelling underneath the shop-houses, the grouting pressure on both machines were maintained at 450 kPa and the grout volume was set to be 2.5m^3 per ring as compared a theoretical volume of 1.9m^3 as shown in Figure 5.

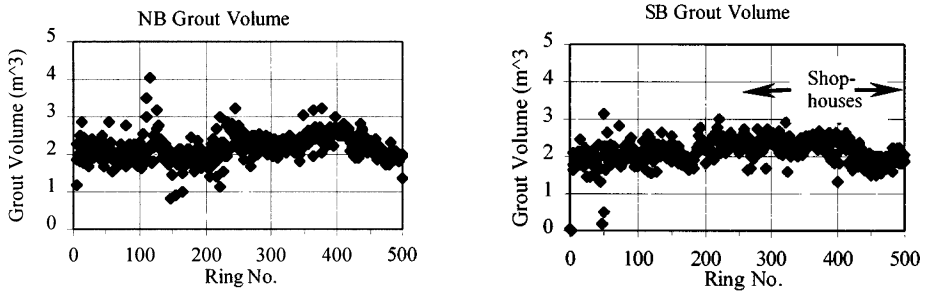


Figure 5. Grout volume.

4 MONITORING RESULTS

The settlement troughs measured at HI1 and MI4 are shown in Figure 6 and Figure 7, respectively.

The settlements at the two arrays were kept less than 5mm when the TBM passed them. In array HI1 some heave may have been generated when the SB TBM passed it. The settlements however continued after the TBMs have passed the monitoring arrays. The maximum settlement measured at the NB sections of the arrays was 7mm. On the SB section, the maximum settlement was 10mm. It is believed that consolidation of the marine clay above the tunnel crown is the main contribution factor for the continuing settlements.

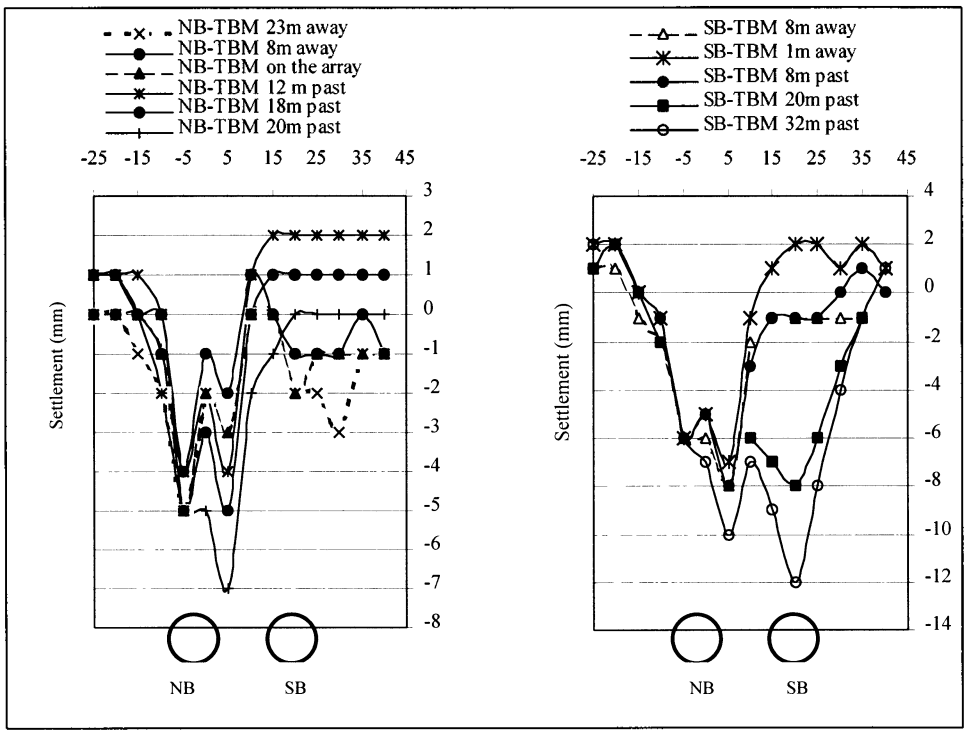


Figure 6. Settlement at HI1

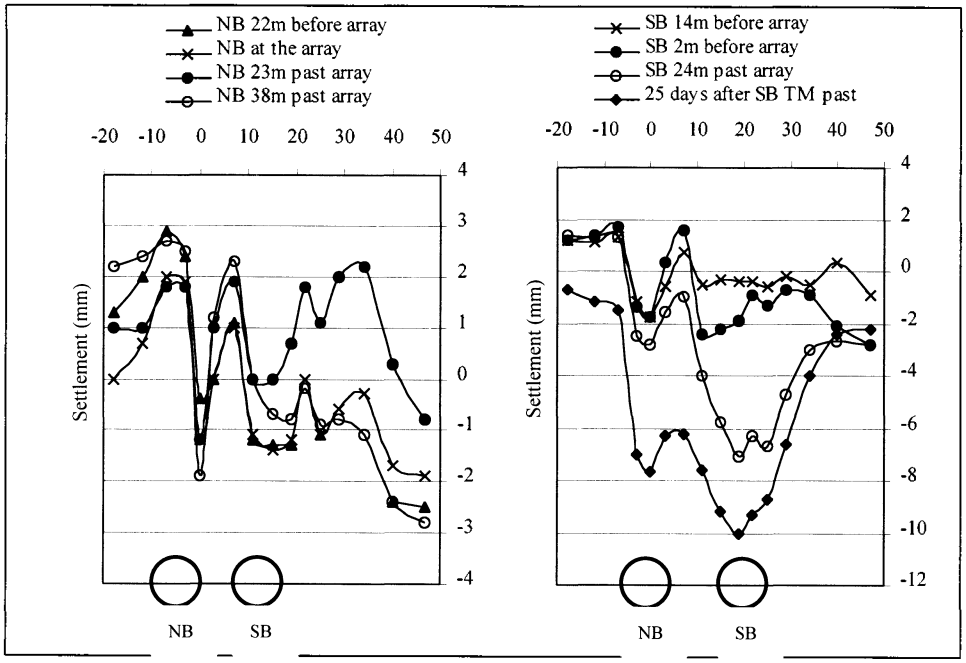


Figure 7. Settlement at MI4

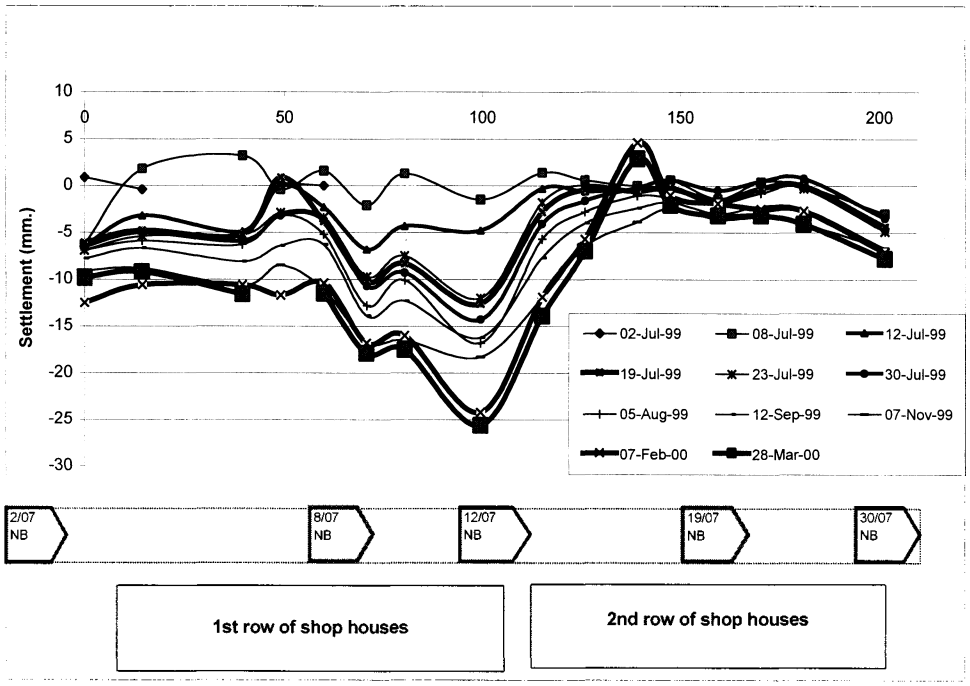


Figure 8. Longitudinal settlements along centre line of NB tunnel

The longitudinal settlement along the centre line of the NB tunnel is shown in Figure 8. High face pressures have resulted in consolidation settlements under the shophouses. The maximum settlement was not more than 30mm. The dip in the settlement corresponds to the occurrence of a pocket of loose sands as shown in Figure 2.

5 CONCLUSIONS

24 hour monitoring of surface instrumentation and monitoring of the EPBM's operational parameters have assisted in the successful completion of the tunnelling works beneath the shophouses without incident.

High face pressures above the overburden pressure were applied to minimise the settlement of the shophouses during excavation. However this resulted in consolidation settlements under the shophouses.

Close control of grouting was implemented to ensure that the gap around the lining was immediately filled to minimize settlement above the EPBMs. The grout volume was generally higher than the theoretical volume of the annulus with a grout pressure higher than the overburden pressure.